Notes Toward the Goreinov–Tyrtyshnikov–Zamarashkin Hypothesis in the First Open Case

Draft template for the case $(n, k) = (6, 3)$

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Abstract

This is a working template for a proof attempt in the first open case of the Goreinov–Tyrtyshnikov–Zamarashkin row-selection hypothesis. The target is the statement that every real 6×3 matrix with orthonormal columns contains a 3×3 row submatrix whose least singular value is at least $1/\sqrt{6}$.

The document deliberately separates proved reductions and structural lemmas from conjectural or numerical material. It records: the formulation in terms of principal submatrices of a rank-three orthoprojector; the single-pair extension mechanism; the equal-leverage slice and its involution model; two obstructions to natural proof strategies; and a minimax/KKT reformulation that appears to be the most promising remaining route.

Contents

1 The target

Let

$$\mathrm{St}(n, k) = \{A \in \mathbb{R}^{n\times k} : A^T A = I_k\}.$$

For $A \in \mathrm{St}(n, k)$ write its rows as $r_1, \dots, r_n \in \mathbb{R}^k$, set

$$P = AA^T, \qquad \ell_i = P_{ii} = \|r_i\|^2, \qquad c_{ij} = P_{ij} = \langle r_i, r_j\rangle.$$

Thus P is a rank-k orthogonal projector. For $I \subset [n]$ let P_{II} be the corresponding principal submatrix. If $|I| = k$, then

$$P_{II} = A_I A_I^T,$$

so the eigenvalues of P_{II} are the squared singular values of A_I.

Conjecture 1.1 (GTZ in the first open case)**.** *For every $A \in \mathrm{St}(6,3)$ there is a triple $I \subset [6]$ such that*

$$\lambda_{\min}(P_{II}) \geq \frac{1}{6}.$$

Equivalently, $\sigma_{\min}(A_I) \geq 1/\sqrt{6}$ for some 3×3 row submatrix.

It is useful to write

$$F(A) = \max_{|I|=3} \lambda_{\min}(P_{II}), \qquad f(A) = \max_{|I|=3} \sigma_{\min}(A_I).$$

Then $F(A) = f(A)^2$, and Conjecture 1.1 is

$$\min_{A \in \mathrm{St}(6,3)} F(A) \geq \frac{1}{6}.$$

2 Small leverage reduction

The standard Case A reduction works for all k. In the present case it says that rows with leverage at most $1/6$ can be removed at no loss, provided the smaller instance is known.

Proposition 2.1 (Case A reduction, specialized)**.** *Let $A \in \mathrm{St}(6,3)$ and suppose some row has $\ell_i \leq 1/6$. Rotate columns so that this row is $(b,0,0)$, where $b^2 = \ell_i$, delete the row, and renormalize the first column by $(1-\ell_i)^{-1/2}$. This gives a matrix $\widetilde{B} \in \mathrm{St}(5,3)$. For every triple J of rows of $\widetilde{B}$, with corresponding triple $\varphi(J) \subset [6] \setminus \{i\}$,*

$$\sqrt{1-\ell_i}\,\sigma_{\min}(\widetilde{B}_J) \leq \sigma_{\min}(A_{\varphi(J)}) \leq \sigma_{\min}(\widetilde{B}_J).$$

Consequently, if the GTZ bound is known for $(5,3)$, then it holds for every $A \in \mathrm{St}(6,3)$ with a row of leverage at most $1/6$.

Proof. After the rotation and deletion, the remaining columns have Gram matrix $\mathrm{diag}(1-\ell_i, 1, 1)$. Multiplication of the first column by $(1-\ell_i)^{-1/2}$ restores orthonormality. Pulling a triple back multiplies on the right by $\mathrm{diag}(\sqrt{1-\ell_i}, 1, 1)$, and the displayed singular value bounds follow from the min–max principle. If $\sigma_{\min}(\widetilde{B}_J) \geq 1/\sqrt{5}$, then

$$\sigma_{\min}(A_{\varphi(J)}) \geq \sqrt{1-\ell_i}\,\frac{1}{\sqrt{5}} \geq \sqrt{\frac{5}{6}}\,\frac{1}{\sqrt{5}} = \frac{1}{\sqrt{6}}.$$

$\square$

Remark 2.2. This reduction is useful but does not by itself settle $(6,3)$, since it uses the neighboring case $(5,3)$. By spectral duality, $(5,3)$ is equivalent to $(5,2)$, which is covered by the known $k=2$ theorem. Thus the genuinely new part of $(6,3)$ may be restricted to the core region

$$\ell_i > 1/6 \quad \text{for every } i.$$

3 The single-pair extension mechanism

Fix $A \in \mathrm{St}(6,3)$ and a pair $\{i,j\}$. Let

$$\Gamma = P_{\{i,j\}\{i,j\}} = \begin{pmatrix} \ell_i & c_{ij} \\ c_{ij} & \ell_j \end{pmatrix}, \qquad \mathrm{spec}(\Gamma) = \{\mu_1, \mu_2\}, \quad \mu_1 \geq \mu_2.$$

For $p \notin \{i,j\}$ set

$$v_p = (c_{ip}, c_{jp})^T.$$

Assume $\mu_2 > 1/6$, so $\Gamma - \frac{1}{6}I_2$ is positive definite, and define

$$q(p) = \ell_p - \frac{1}{6} - v_p^T \left(\Gamma - \frac{1}{6}I_2\right)^{-1} v_p.$$

Lemma 3.1 (Schur criterion)**.** *For $p \notin \{i,j\}$,*

$$\lambda_{\min}(P_{\{i,j,p\},\{i,j,p\}}) \geq \frac{1}{6} \quad \iff \quad q(p) \geq 0.$$

Proof. Subtract $\frac{1}{6}I_3$ from the triple Gram matrix and take the Schur complement of the positive block $\Gamma - \frac{1}{6}I_2$. □

Define, for $\mu > 1/6$,

$$h(\mu) = \frac{1-\mu}{6\mu - 1}.$$

Lemma 3.2 (Extension lemma)**.** *If a pair satisfies*

$$\mu_2 > \frac{1}{6}, \qquad h(\mu_1) + h(\mu_2) \leq \frac{1}{3},$$

then some triple containing that pair is $1/6$-good.

Proof. The moment identity

$$\sum_{p \notin \{i,j\}} v_p v_p^T = \Gamma - \Gamma^2$$

and $\sum_p \ell_p = 3$ give

$$\sum_{p \notin \{i,j\}} q(p) = \frac{7}{3} - \frac{5\mu_1}{6\mu_1 - 1} - \frac{5\mu_2}{6\mu_2 - 1} = \frac{1}{3} - h(\mu_1) - h(\mu_2).$$

Thus the assumed inequality makes the sum of the four Schur slacks nonnegative. At least one $q(p)$ is nonnegative, and Lemma 3.1 applies. □

Corollary 3.3 (A trace branch)**.** *If some pair satisfies*

$$\ell_i + \ell_j \geq \frac{13}{9},$$

then Conjecture 1.1 holds for A.

Proof. Let $T = \ell_i + \ell_j = \mu_1 + \mu_2$. Since $\mu_1 \leq 1$, the inequality $T \geq 13/9$ implies $\mu_2 \geq 4/9 > 1/6$. The function h is decreasing and convex on $(1/6, \infty)$. The maximum of $h(\mu_1) + h(13/9 - \mu_1)$ on the admissible interval occurs at an endpoint, and the endpoint values give at most $1/3$. Lemma 3.2 finishes the proof. □

4 The equal-leverage slice

The slice

$$\ell_1 = \cdots = \ell_6 = \frac{1}{2}$$

has a particularly rigid form. Put

$$J = 2P - I_6.$$

Then J is a symmetric orthogonal involution with zero diagonal and

$$\operatorname{spec}(J) = \{+1, +1, +1, -1, -1, -1\}.$$

Conversely, any such J gives a rank-three projector $P = (I + J)/2$ with diagonal entries $1/2$.

Write

$$a_{ij} = J_{ij} \quad (i \neq j).$$

Equivalently, after setting $u_i = \sqrt{2}\, r_i$, the u_i are unit vectors in $\mathbb{R}^3$ with

$$\sum_{i=1}^{6} u_i u_i^T = 2I_3, \qquad a_{ij} = \langle u_i, u_j \rangle.$$

The involution identity gives the useful row and flow equations

$$\sum_{k \neq i} a_{ik}^2 = 1, \qquad \sum_{m \notin \{i,j\}} a_{im} a_{jm} = 0.$$

Lemma 4.1 (Slice criterion for a good triple)**.** *For a triple $T = \{i, j, k\}$ set*

$$p_T = a_{ij}^2 + a_{ik}^2 + a_{jk}^2, \qquad q_T = a_{ij} a_{ik} a_{jk}.$$

Then T is $1/6$-good if and only if

$$q_T \geq \frac{p_T}{3} - \frac{4}{27}.$$

Proof. On the slice,

$$P_{TT} = \frac{1}{2}(I_3 + J_{TT}).$$

Thus T is $1/6$-good iff $\lambda_{\min}(J_{TT}) \geq -2/3$. The zero-diagonal 3×3 matrix J_{TT} has characteristic polynomial

$$x^3 - p_T x - 2q_T.$$

Since J_{TT} is a principal submatrix of an involution, its eigenvalues lie in $[-1, 1]$ and have sum zero, so at most one can be below $-2/3$. Therefore the sign of $\det(J_{TT} + \frac{2}{3} I_3)$ detects whether the least eigenvalue is at least $-2/3$. But

$$\det(J_{TT} + \tfrac{2}{3} I_3) = \frac{8}{27} - \frac{2}{3} p_T + 2q_T,$$

which gives the criterion. $\square$

The Veronese reformulation is also useful. Let

$$w_i = u_i u_i^T - \frac{1}{3} I_3 \in \operatorname{Sym}_0(\mathbb{R}^3).$$

Then $\sum_i w_i = 0$ and, for a triple $T = \{i, j, k\}$,

$$\tau_T = \operatorname{tr}(w_i w_j w_k) = q_T - \frac{p_T}{3} + \frac{2}{9}.$$

Thus Lemma 4.1 is equivalent to

$$T \text{ is good} \quad \Longleftrightarrow \quad \tau_T \geq \frac{2}{27}.$$

Moreover,

$$\sum_{|T|=3} \tau_T = \frac{4}{9}, \qquad \sum_{|T|=3} p_T = 12, \qquad \sum_{|T|=3} q_T = 0.$$

Remark 4.2 (Why averaging alone fails)**.** The mean value of τ_T over the 20 triples is $1/45$, which is below the required threshold $2/27$. Therefore the slice theorem, if true, must use more than unsigned averaging of the τ_T.

5 Pinned block spectra

The involution model also gives exact spectra for large principal blocks. These facts are potential constraints in an active-set or algebraic proof.

Lemma 5.1 (Five-by-five blocks)**.** *Let J be any symmetric involution with*

$$\operatorname{spec}(J) = \{+1, +1, +1, -1, -1, -1\},$$

not necessarily with zero diagonal. If $F = [6] \setminus \{v\}$ and $d_v = J_{vv}$, then

$$\operatorname{spec}(J_{FF}) = \{1, 1, -d_v, -1, -1\}.$$

Proof. This follows from Cauchy interlacing. The first two and last two eigenvalues are pinned at $1, 1$ and $-1, -1$, and the remaining eigenvalue is determined by the trace. □

Lemma 5.2 (Four-by-four blocks on the slice)**.** *Assume J lies on the slice. If $F = [6] \setminus \{v, w\}$, then*

$$\operatorname{spec}(J_{FF}) = \{1, t, -t, -1\}, \qquad t = |a_{vw}|.$$

Proof. Interlacing pins one eigenvalue at 1 and one at -1. The trace of J_{FF} is zero on the slice, so the remaining two eigenvalues sum to zero. Their squared sum is fixed by the Frobenius norm of J_{FF}, which is controlled by the row-mass identities and equals $2t^2$ for the remaining pair. □

6 Known obstructions

The following observations are intended as guardrails for the proof attempt.

Proposition 6.1 (Single-pair boundary obstruction; informal statement)**.** *There is no universal $\delta_0 > 0$ such that every qualifying pair in* $\mathrm{St}(6, 3)$ *with*

$$0 < h(\mu_1) + h(\mu_2) - \frac{1}{3} \leq \delta_0$$

extends to a $1/6$-good triple through that pair.

Proof status. This is a recorded exact-arithmetic obstruction in the accompanying notes. It is obtained by perturbing a Nesterenko extremal along a Grassmann tangent direction. The selected pair moves into positive excess, all four triples through it move below the threshold to first order, while an external active triple moves upward and preserves the global GTZ value locally. □

Remark 6.2. Proposition 6.1 says that Lemma 3.2 is essentially the maximal reach of any proof using only one pair and its second-moment data.

Proposition 6.3 (The icosahedral obstruction)**.** *On the equal-leverage icosahedral frame, every pair satisfies*

$$h(\mu_1) + h(\mu_2) = \frac{13}{11} > \frac{1}{3},$$

so no pair lies in the extension branch of Lemma 3.2. However good triples exist; indeed the coherent triples are precisely the good triples.

Proof. For the icosahedral equiangular tight frame, $\ell_i = 1/2$ and $c_{ij}^2 = 1/20$ for all $i \neq j$. Hence each pair has eigenvalues

$$\mu_{1,2} = \frac{1}{2} \pm \frac{1}{\sqrt{20}},$$

and a direct substitution gives the displayed h-sum. For a triple, the off-diagonal signs can be switched to either all positive or all negative according to the sign of $c_{ij}c_{ik}c_{jk}$. The all-positive case has least eigenvalue $1/2 - 1/\sqrt{20} > 1/6$, while the all-negative case has least eigenvalue $1/2 - 2/\sqrt{20} < 1/6$. □

7 A minimax/KKT reformulation

The obstruction above suggests replacing a single-pair view by the active set of triples for the nonsmooth objective

$$F(A) = \max_{|T|=3} \lambda_{\min}(P_{TT}).$$

Let $\mathcal{A}(A)$ denote the active triples at A:

$$\mathcal{A}(A) = \{T : \lambda_{\min}(P_{TT}) = F(A)\}.$$

Definition 7.1. A point $A \in \mathrm{St}(6,3)$ is called a KKT point of F if it is Clarke-stationary for the maximum of the active eigenvalue functions. At points where the active least eigenvalues are simple, this means

$$0 \in \mathrm{conv}\{\nabla\lambda_{\min}(P_{TT}) : T \in \mathcal{A}(A)\} \subset T_A^* \mathrm{St}(6,3),$$

with gradients projected to the Stiefel tangent space.

Proposition 7.2 (KKT reduction)**.** *Conjecture 1.1 is equivalent to the assertion that no KKT point of F has value below $1/6$.*

Proof. Since $\mathrm{St}(6,3)$ is compact and F is continuous, F attains a global minimum. Every global minimizer is KKT/Clarke-stationary. Hence if no KKT point has value below $1/6$, the global minimum is at least $1/6$. The converse is immediate. □

Remark 7.3 (Local extremals)**.** Numerical evidence in the notes suggests that the known equality examples are strict local minima of F: their active triple gradients positively span the tangent dual. A useful finite task is to turn this into exact certificates, one equality type at a time.

8 A possible route for the proof

The observations above suggest the following work plan.

Step 1. Use Case A and duality to restrict attention to the core region $\ell_i > 1/6$.

Step 2. Prove the equal-leverage slice theorem: for every zero-diagonal symmetric orthogonal J with spectrum $(+1)^3(-1)^3$, some triple satisfies the criterion of Lemma 4.1. The current candidate extremal inside the slice is the pentagonal matching configuration, which appears to have margin about 0.0394 above 1/6.

Step 3. Prove exact local KKT certificates at the known equality configurations. This should be a finite algebraic problem: compute the active gradients and exhibit an exact positive convex dependence.

Step 4. Enumerate possible active-set types for a hypothetical KKT point with $F < 1/6$ and rule them out, using the pinned block spectra, complementarity, and the Schur slack identities.

Step 5. If the previous step becomes too large, set up a symmetry-reduced SOS certificate with the equality manifold imposed as equations. Low-degree generic relaxations are expected to be too weak because averaging alone is too weak on the slice.

9 Checklist of statements and status

Statement	Current status
Case A reduction	proved
Schur criterion and extension lemma	proved
Trace branch $\ell_i + \ell_j \geq 13/9$	proved
Equal-leverage involution model	proved
Slice triple criterion	proved
Pinned 5×5 and 4×4 spectra	proved
Single-pair boundary obstruction	proved in accompanying notes
Icosahedral obstruction and coherent rescue	proved
Equal-leverage slice theorem	open / numerically supported
Exact KKT positive spanning at equality examples	numerical evidence
No KKT point below 1/6	open
Full Conjecture 1.1	open

References

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